

# I. Poisson Regression (Lecture 12)

**Poisson Log-Linear Model (canonical log link):**

$$\log \mu_i = \mathbf{x}_i^T \boldsymbol{\beta}, \quad \mu_i = \exp(\mathbf{x}_i^T \boldsymbol{\beta})$$

- $\beta_j$ : change in **log mean count** per unit of  $x_j$ .
- $\exp(\beta_j)$ : **rate ratio** (multiplicative effect on mean count).

**Goodness of Fit** ( $\chi^2_{n-p}$  when  $\mu_i$  large):

- **Deviance:**  $D = 2 \sum \{y_i \log(y_i/\hat{\mu}_i) - (y_i - \hat{\mu}_i)\}$
- **Pearson  $\chi^2$ :**  $G = X^2 = \sum (y_i - \hat{\mu}_i)^2 / \hat{\mu}_i$

**Poisson Rate & Offset:** When events occur over exposure  $n_i$  at rate  $\lambda_i$ :

$$E(Y_i) = \mu_i = n_i \lambda_i \Rightarrow \log \mu_i = \log n_i + \mathbf{x}_i^T \boldsymbol{\beta}$$

- Offset is **fixed** (coefficient = 1, not estimated).

# II. Overdispersion (Lecture 13)

**Definition:**  $\text{Var}(Y) > E(Y)$ , i.e.,  $\text{Var}(Y) = \phi \mu$  with  $\phi > 1$ .

**Diagnostics:**

- **Half-normal plot:** Order  $|r_{(i)}|$ , plot vs  $\Phi^{-1}(\frac{n+i+0.5}{2n+1.125})$ . Reference  $y = x$ . Slope  $\approx \sqrt{\phi}$ . Constant overdispersion  $\Rightarrow$  linear deviation.

**Method 1: Quasi-Poisson (Dispersion Parameter):**

$$\hat{\phi} = \frac{G}{n-p} = \frac{1}{n-p} \sum \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i}$$

$$\hat{\beta}_Q = \hat{\beta}_{MLE}, \quad \text{cov}(\hat{\beta}_Q) = \hat{\phi} \cdot \text{cov}(\hat{\beta}_{MLE})$$

**Deviance Analysis (F-test) with overdispersion** for nested  $M_0 \subset M_1$ :

$$F = \frac{(D_0 - D_1)/((p_1 - p_0))}{\hat{\phi}} \sim F_{p_1 - p_0, n - p_1}, \quad \hat{\phi} = G_{M_1}/(n - p_1)$$

**Method 2: Negative Binomial Regression:**

$$E(Y) = \mu, \quad \text{Var}(Y) = \mu + \mu^2/\phi$$

- $\phi \rightarrow \infty \Rightarrow$  Poisson.

# Zero-Inflated Poisson (ZIP)

When data have **excessive zeros**:

- Latent  $Z_i$  (binary):  $P(Z_i = 0) = \pi_i$  (always zero).
- $Y_i | Z_i = 0 \equiv 0$ ;  $Y_i | Z_i = 1 \sim \text{Poisson}(\lambda_i)$ .
- Model:  $\log(\lambda_i) = \mathbf{x}_i \boldsymbol{\beta}$  and  $\text{logit}(\pi_i) = \mathbf{z}_i \boldsymbol{\gamma}$  (different/same predictors).
- $P(Y = 0) = \pi + (1 - \pi)e^{-\lambda}$ .

# III. LDA: Overview (Lectures 15-20)

**Three Modeling Approaches:**

- **Marginal:**  $E(Y_{ij}) = \mathbf{X}_{ij}^T \boldsymbol{\beta}$ ,  $\text{Var}(\mathbf{Y}_i) = \boldsymbol{\Sigma}_i$ .
- **Mixed effects:**  $E(Y_{ij} | \boldsymbol{\beta}_i) = \mathbf{X}_{ij}^T \boldsymbol{\beta}_i$ ,  $\boldsymbol{\beta}_i = \boldsymbol{\beta} + \mathbf{U}_i$ .
- **Transition:**  $E(Y_{ij} | Y_{i,j-1}, \dots, Y_{i1}, \mathbf{X}_{ij})$ .

# Covariance Pattern Models - GLS (Lecture 16)

(a) **Compound Symmetry / Exchangeable:**  $\text{Var} = \sigma^2$ ,  $\text{corr} = \rho$  (any pair). 2 params. Strong assumption.

(b) **Toeplitz:**  $\text{corr}(Y_{ij}, Y_{i,j+k}) = \rho_k$ .  $n$  params. Equal time intervals only.

(c) **Autoregressive AR(1):** Special Toeplitz:  $\rho_k = \rho^k$ . **2 params** total. Equal intervals.

(d) **Banded:**  $\text{corr}=0$  beyond an interval.

(e) **Exponential:**  $\text{corr}(Y_{ij}, Y_{ik}) = \rho^{|t_j - t_k|}$ . AR(1) generalization for **uneven** times.

(f) **Unstructured:** All variances/covariances free. Most flexible, most params.

# Linear Mixed Effects - LMM (Lecture 17)

**General LMM:**  $\mathbf{Y}_i = \mathbf{X}_i \boldsymbol{\beta} + \mathbf{Z}_i \mathbf{b}_i + \boldsymbol{\epsilon}_i$ ,  $i = 1, \dots, m$ .

- $\boldsymbol{\beta}$ : fixed effects ( $p \times 1$ ).  $\mathbf{b}_i$ : random effects ( $q \times 1$ ).
- $\mathbf{b}_i \sim N(\mathbf{0}, \mathbf{G})$ ,  $\boldsymbol{\epsilon}_i \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$ , indep.
- **Conditional mean:**  $E(\mathbf{Y}_i | \mathbf{b}_i) = \mathbf{X}_i \boldsymbol{\beta} + \mathbf{Z}_i \mathbf{b}_i$ .
- **Marginal mean:**  $E(\mathbf{Y}_i) = \mathbf{X}_i \boldsymbol{\beta}$  (population).
- **Marginal Var:**  $\text{Var}(\mathbf{Y}_i) = \mathbf{Z}_i \mathbf{G} \mathbf{Z}_i^T + \sigma^2 \mathbf{I}$ .

**Random Intercept Model**

$$Y_{ij} = \beta_0 + \mathbf{X}_{ij}^T \boldsymbol{\beta} + b_i + \epsilon_{ij}, \quad b_i \sim N(0, \sigma_b^2).$$

- $\text{Var}(Y_{ij}) = \sigma_b^2 + \sigma^2$ .
- $\text{Cov}(Y_{ij}, Y_{ik}) = \sigma_b^2$  (induces **compound symmetry**).
- **ICC:**  $\rho = \sigma_b^2 / (\sigma_b^2 + \sigma^2)$ .

**Random Intercept and Slope Model**

$$Y_{ij} = \beta_1 + \beta_2 t_{ij} + b_{1i} + b_{2i} t_{ij} + \epsilon_{ij}$$

$$\mathbf{b}_i = (b_{1i}, b_{2i})^T \sim N(\mathbf{0}, \mathbf{G}), \quad \mathbf{G} = \begin{pmatrix} g_{11} & g_{12} \\ g_{12} & g_{22} \end{pmatrix}.$$

- $\text{Var}(Y_{ij}) = g_{11} + 2t_{ij}g_{12} + t_{ij}^2g_{22} + \sigma^2$  (varies w/ time).
- $\text{Cov}(Y_{ij}, Y_{ik}) = g_{11} + (t_{ij} + t_{ik})g_{12} + t_{ij}t_{ik}g_{22}$ .

# GEE (Lecture 19)

**Marginal Model for non-Gaussian longitudinal:**

$$g(\mu_{ij}) = \mathbf{X}_{ij}^T \boldsymbol{\beta}, \quad \text{Var}(Y_{ij}) = \phi V(\mu_{ij}), \quad \text{corr}(Y_{ij}, Y_{ik}) = \alpha_{jk}$$

**Interpretation:**  $\boldsymbol{\beta}$  has **population-averaged** meaning. e.g., logistic GEE:  $\beta_j = \log \text{OR}$  comparing **subgroups** of population differing in  $X_j$ .

# GLMM (Lecture 20)

**Generalized Linear Mixed Model:** Conditional on  $\mathbf{b}_i$ :

$$Y_{ij} | \mathbf{b}_i \sim \text{EF}, \quad g(E(Y_{ij} | \mathbf{b}_i)) = \mathbf{X}_{ij}^T \boldsymbol{\beta} + \mathbf{Z}_{ij}^T \mathbf{b}_i$$

$$\mathbf{b}_i \sim N(\mathbf{0}, \mathbf{G}).$$

**Examples:**

- **Logistic GLMM (random intercept):**

$$\text{logit}(E(Y_{ij} | b_i)) = b_i + \beta_1 + X_{ij} \beta_2, \quad b_i \sim N(0, \sigma_b^2)$$

- **Poisson GLMM (random int+slope):**

$$\log E(Y_{ij} | \mathbf{b}_i) = (b_{1i} + \beta_1) + (b_{2i} + \beta_2) t_{ij}$$

## Marginal vs GLMM (KEY DIFFERENCE!)

$$E(\mathbf{Y}_i) = \int g^{-1}(\mathbf{X}_i\beta + \mathbf{Z}_i\mathbf{b}_i)f(\mathbf{b}_i)d\mathbf{b}_i \neq g^{-1}(\mathbf{X}_i\beta_*)$$

- For **nonlinear**  $g$  (logit/log), GLMM  $\beta \neq$  Marginal  $\beta_*$ .
- For **linear**  $g$  (Gaussian), they coincide.
- **Marginal**  $\beta$ : population-averaged effect.
- **GLMM**  $\beta$ : subject-specific (conditional on  $\mathbf{b}_i$ ) effect — for **same subject**, log-OR per unit  $X$  change.
- $|\beta_{GLMM}| > |\beta_{Marginal}|$  usually for binary outcomes.

## IV. Survival Analysis (Lecture 21)

- **CDF**:  $F(t) = P(T \leq t)$ .
- **Survival**:  $S(t) = P(T > t) = 1 - F(t)$ .  $S(0) = 1$ ,  $S(\infty) = 0$ , decreasing.
- **Hazard rate**:  $h(t) = \frac{f(t)}{S(t)} = -\frac{d \log S(t)}{dt}$
- **Cumulative hazard**:  $H(t) = \int_0^t h(\tau) d\tau$ .

**Key relations**:  $S(t) = \exp\{-H(t)\}$ ,  $f(t) = h(t)S(t)$ .

**Parametric distributions**:

- **Exponential**:  $f(t) = \lambda e^{-\lambda t}$ ,  $S(t) = e^{-\lambda t}$ ,  $h(t) = \lambda$  (constant!).
- **Weibull**:  $S(t) = e^{-\lambda t^\alpha}$ ,  $h(t) = \lambda \alpha t^{\alpha-1}$ ,  $f(t) = \lambda \alpha t^{\alpha-1} e^{-\lambda t^\alpha}$ .  
 $-\alpha > 1$ : increasing hazard;  $\alpha < 1$ : decreasing;  $\alpha = 1$ : exponential.

## Kaplan-Meier Estimator (Nonparametric)

For ordered times  $t_1 < \dots < t_m$ :

$$\hat{S}(t) = \prod_{i:t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

- $n_i$ : # at risk just **before**  $t_i$ .  $d_i$ : # events at  $t_i$ . Censored at  $t_i$  counted in  $n_i$  but not  $d_i$ .
- $\hat{\lambda}_i = d_i/n_i$  = conditional prob of death at  $t_i$ .

**Greenwood Formula (variance of KM)**:

$$\widehat{\text{Var}}(\hat{S}(t)) = \hat{S}(t)^2 \sum_{i:t_i \leq t} \frac{d_i}{(n_i - d_i)n_i}$$

95% CI:  $\hat{S}(t) \pm 1.96\widehat{SE}$  (truncate to  $[0,1]$ ). Better: build CI for  $\log S$  or  $\log(-\log S)$  then transform.

**Cumulative hazard**:

- KM-based:  $\hat{H}(t) = -\log \hat{S}(t)$ .
- **Nelson-Aalen** (better):  $\tilde{H}(t) = \sum_{t_i \leq t} d_i/n_i$  with  $\widehat{\text{Var}}(\tilde{H}) = \sum d_i/n_i^2$ .
- **Fleming-Harrington**:  $\hat{S}(t) = \exp(-\tilde{H}(t))$ .

## V. Survival Curves (Lecture 22)

**Complete data, fixed time**:  $\chi^2$  or Fisher exact on  $2 \times 2$  (failed before  $t$ ).

**Censored data, fixed time  $t$**  (use KM at  $t$ ):

$$T = \frac{\hat{S}_A(t) - \hat{S}_B(t)}{\widehat{SD}\{\hat{S}_A(t) - \hat{S}_B(t)\}} \sim N(0, 1)$$

$$\widehat{\text{Var}}\{\hat{S}_A - \hat{S}_B\} = \widehat{\text{Var}}\hat{S}_A + \widehat{\text{Var}}\hat{S}_B$$

## Log-Rank Test (Overall Difference)

$H_0 : S_A(t) = S_B(t) \forall t$  (equivalently  $h_A = h_B$ ).

At each **uncensored** time  $t_i$  (pooled): build  $2 \times 2$  (A vs B, event vs no event).

- $d_{A,i} \sim \text{Hypergeometric}(n_i, m_{D,i}, n_{A,i})$  under  $H_0$ .
- $E(d_{A,i}) = m_{D,i}n_{A,i}/n_i$ .
- $\text{Var}(d_{A,i}) = \frac{n_{A,i}n_{B,i}m_{D,i}m_{\bar{D},i}}{n_i^2(n_i - 1)}$ .

**Test Statistic**:

$$Z = \frac{\sum_i \{d_{A,i} - E(d_{A,i})\}}{\sqrt{\sum_i \text{Var}(d_{A,i})}} \sim N(0, 1), \quad Z^2 \sim \chi_1^2$$

**Weighted Log-Rank**:  $T = \sum w_i \{d_{A,i} - E(d_{A,i})\}$ .

- $w_i = 1$ : log-rank (most powerful when HR proportional/late).
- $w_i = n_i$ : **Gehan**.  $w_i = \sqrt{n_i}$ : **Tarone-Ware**.  $w_i = \hat{S}(t_i)^\rho \{1 - \hat{S}(t_i)\}^\gamma$ : **Fleming-Harrington**.

## VI. Survival Regression (Lecture 23)

**(a) Accelerated Failure Time (AFT) — Parametric**

$$\log T_i = \mathbf{X}_i^T \beta + \epsilon_i, \quad T_i = \exp(\mathbf{X}_i^T \beta) T_{0i}$$

- $\epsilon_i$  has parametric distribution (Weibull/Exponential/Log-normal).
- $\exp(\beta_j)$ : **time accelerator**. With unit  $\uparrow x_j$ , expected event time is  $\exp(\beta_j) \times$  original.

**(b) Cox Proportional Hazards (PH) — Semi-parametric**

$$h_i(t) = h_0(t) \exp(\mathbf{X}_i^T \beta)$$

- $h_0(t)$ : **baseline hazard**, unspecified (nonparametric).
- $\exp(\beta)$ : **hazard ratio** (parametric).
- **NO intercept** in  $\mathbf{X}^T \beta$  (absorbed in  $h_0$ ).
- Hazard ratio  $h_i(t)/h_k(t) = \exp((\mathbf{X}_i - \mathbf{X}_k)^T \beta)$  — **constant over  $t$** .
- $\beta_j$ : log HR per unit  $\uparrow x_j$ .

**(c) Time-Dependent Covariates**

$h_i(t) = h_0(t) \exp(\mathbf{x}_i(t)^T \beta)$ . Use partial likelihood with  $\mathbf{x}_j$  replaced by  $\mathbf{x}_j(t_{i'})$  in risk set.  $\beta_j$ : log HR per unit change at a given time.

**(d) Stratified PH Model**

PH may not hold across all covariates. If a stratification covariate  $S$  violates PH:

$$h_{is}(t) = h_{0s}(t) \exp(\mathbf{x}_i^T \beta)$$

- Different baseline hazard per stratum, **same  $\beta$**  (HR constant **within** strata).